

P-EDCA under light load: best P99 parameters vs saturated load

802.11be, nSta=30, CWds×QSRC×PSRC sweep | offered load 0.1 / 0.5 / 1.0 Mbps per STA | P-EDCA STA P99

The three load regimes (EDCA-only VO P99 sets the scene)

■ 0.1 Mbps — uncongested

EDCA-only P99 = 0.92 ms (sub-ms). Nothing for P-EDCA to fix.

■ 0.5 Mbps — congested, not saturated

EDCA-only P99 = 8.25 ms. Real queueing but the priority path can drain it.

■ 1.0 Mbps — saturated

EDCA-only P99 = 13.56 ms. Collision-feedback regime.

0.1 Mbps: ≈ no-op

P-EDCA cuts P99 only +5–6%

best 0.86 ms vs EDCA 0.92 ms — any combo works

0.5 Mbps: sweet spot

+63% P99 (8.2 → 3.0 ms)

QSRC=0, PSRC=3 — aggressive & safe (no tail risk)

Best light-load recipe

QSRC = 0, PSRC = 3, CWds = 0/1

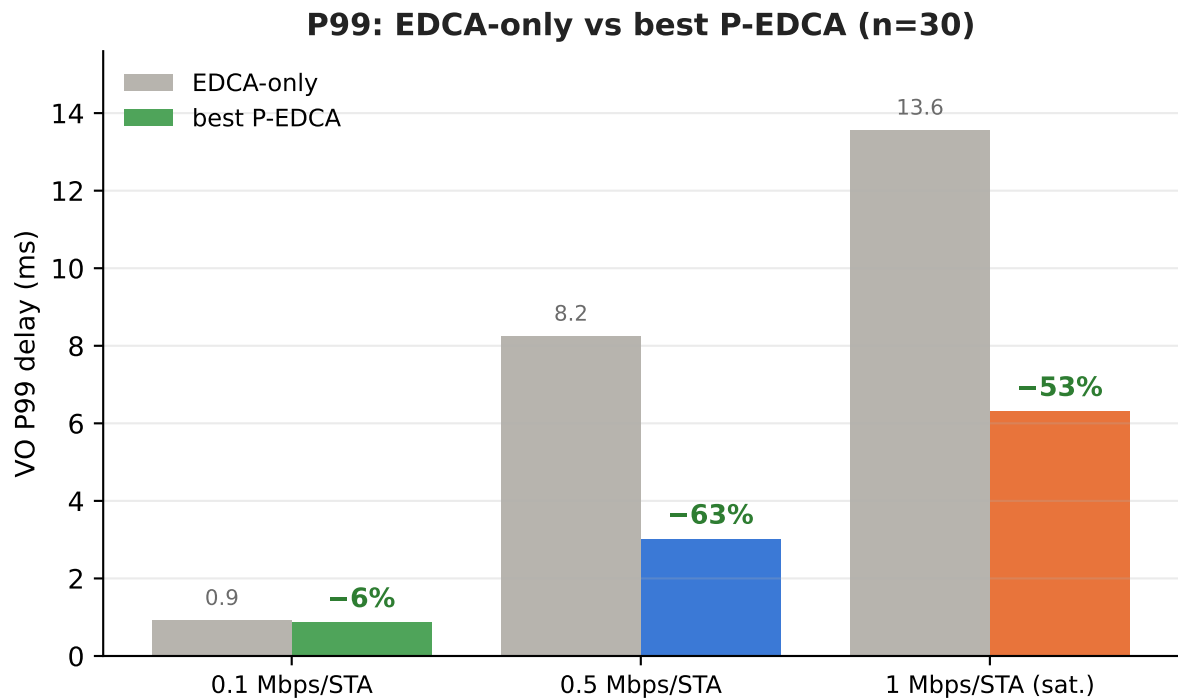
saturated backs QSRC off to 1 (n=5 tail risk)

One-line takeaway

P-EDCA's value tracks how congested EDCA already is: near-zero at 0.1 Mbps, largest at 0.5 Mbps (+60%+), and still large but riskier at saturation. The aggressive recipe (QSRC 0, PSRC 3) wins whenever there is real load to drain.

Best P-EDCA P99 across loads — the gain peaks at moderate load

EDCA-only baseline vs best-of-36-combos P-EDCA (P-EDCA STAs, n=30), with % P99 reduction



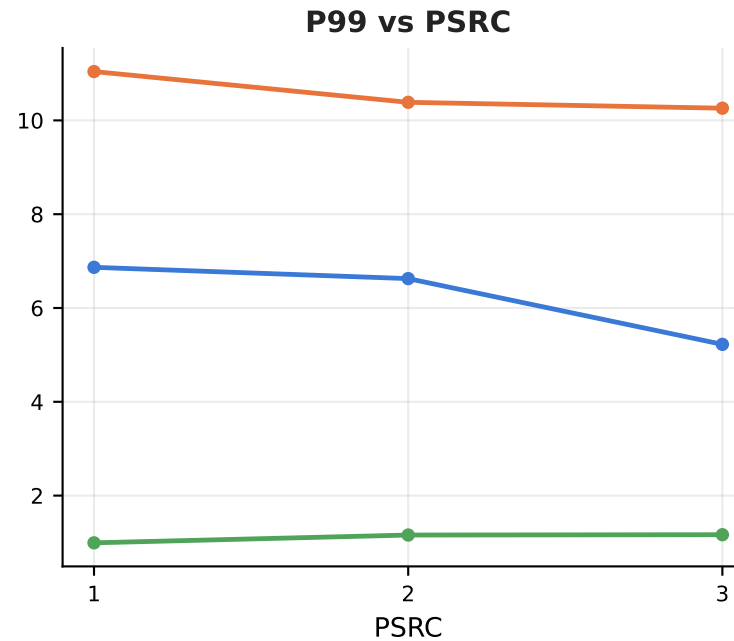
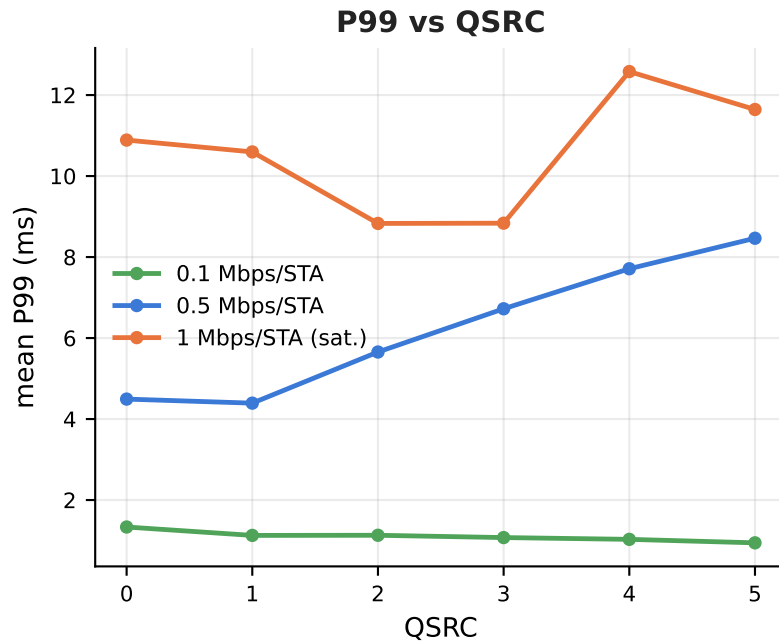
Best combo & P99 per P-EDCA population

Load	nPedca	best c/q/s	P99 (ms)
0.1 Mbps/STA	5	c0 q2 s2	0.87 (-5%)
	15	c0 q5 s1	0.86 (-6%)
	30	c1 q1 s1	0.86 (-6%)
0.5 Mbps/STA	5	c0 q0 s3	3.19 (-61%)
	15	c1 q0 s3	3.30 (-60%)
	30	c1 q0 s3	3.01 (-63%)
1 Mbps/STA (sat.)	5	c0 q1 s3	6.03 (-56%)
	15	c1 q1 s3	5.83 (-57%)
	30	c0 q1 s3	6.31 (-53%)

- Relative gain peaks at moderate load: +5-6% at 0.1 Mbps → +63% at 0.5 Mbps → +53% at saturation. P-EDCA only helps once EDCA itself is congested.
- At 0.1 Mbps the best P99 (0.86 ms) barely beats EDCA (0.92 ms) and the winning combo is inconsistent across nPedca (c0q2s2 / c0q5s1 / c1q1s1) — a sign params don't matter.
- At 0.5 & 1.0 Mbps the winner is consistently aggressive (QSRC 0-1, PSRC 3); the gain is stable across nPedca (5→30), i.e. it does not collapse as P-EDCA penetration rises.

Which knob matters at which load — and the saturation tail-risk

Marginal mean P99 (P-EDCA STAs, n=30) vs QSRC and vs PSRC, per load



Saturation tail-risk

worst-case P99 over 36 combos

0.1: 1.5 ms 0.5: 8.9 ms

1.0 (sat), n=5: 294 ms

aggressive QSRC0+PSRC3 explodes only
at saturation + low penetration

- QSRC: at 0.1 Mbps flat/slightly falling (higher QSRC marginally better — do not trigger P-EDCA when uncongested); at 0.5 Mbps strongly monotonic (q0 best, 4.5 ms vs 8.5 ms at q5); at 1.0 Mbps noisy/non-monotonic.
- PSRC: irrelevant at 0.1 Mbps; clearly larger-is-better at 0.5 Mbps (s3 5.2 ms vs s1 6.9 ms); at saturation PSRC=3 helps on average but is the trigger for the n=5 tail blow-up.
- Robust recommendation — light/moderate (0.1-0.5 Mbps): QSRC=0, PSRC=3, CWds=0/1 (safe, worst-case ≤ 9 ms). Saturated (1.0 Mbps): back QSRC off to 1 to avoid the 294 ms low-penetration tail.